Patrick Iben

Applied mathematics, biostatistics and research engineering

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Profile

Applied mathematician and biostatistician with seven years in clinical biostatistics and twelve years in research. Works to eliminate needless ambiguity, risk, and suffering in decisions that rest on data. Builds statistical frameworks, AI-evaluation tools and public-data systems with traceable inputs and testable assumptions. Independent work includes six research manuscripts, scientific Python software, wearable sensing and 32 public-data consoles. Open to full-time and consulting roles.

Technical expertise

Methods: trial design, estimands, causal inference under interference (spillover and contamination), consilience testing across data sources, simulation, permutation inference, calibration, decision curves, PK/NCA, bioequivalence, safety review, CDISC SDTM/ADaM and statistical analysis plans.

Tools: R (pharmaverse and Shiny), Quarto, SAS, Python, SQL, JavaScript, Git, Docker and LaTeX. Data provenance, automated QC, browser testing, public API integration and numerical validation.

Professional experience

Biostatistician II | Syneos Health | Remote | December 2021 to October 2026

- Served as lead biostatistician from December 2022 on early-phase clinical pharmacology studies, including first-in-human, SAD/MAD, crossover, bioequivalence and bioavailability designs.
- Owned protocol statistical sections, randomization schedules, analysis plans and mock tables, listings and figures; created or reviewed CDISC analysis specifications.
- Performed PK/PD analyses, reviewed outputs and contributed to clinical study reports; reviewed interim study data, annotated CRFs and database designs against analysis requirements.

Associate Biostatistician | Emmes | Rockville, MD | September 2019 to December 2021

- Led and supported federally sponsored Phase 1 and Phase 3 opioid-use-disorder trials; wrote protocol and analysis-plan sections, DSMB reports and final statistical reports.
- Built dynamic study-progress reporting and SAS/R tools; harmonized multi-clinic EHR data and supported external research partners.

Research Assistant and Teaching Assistant | University of Colorado Anschutz | August 2016 to July 2019

- Harmonized hospital databases containing millions of records and hundreds of derived variables for pediatric nursing-cost research and reuse in later studies.
- Advised doctoral nursing candidates on design and sample size; taught graduate biostatistics review sessions and office hours.

Earlier research roles | 2014 to 2016

Postsecondary and Policy Analysis Fellow, State Higher Education Executive Officers; Data Analysis Fellow, Yale School of Child Development; Senior Data Analyst, SCSU Office of Assessment and Planning.

Education

MS, Biostatistics, Data Science Analytics minor | University of Colorado Anschutz | 2019

BS, Mathematics, Applications of Mathematics | Southern Connecticut State University | 2016 | Magna cum laude

Selected projects

Trial biostatistics toolkit and eCRF review apps

A public toolkit for biostatistics consultants: runnable SAS and R templates, dashboards and QC checks for the work a biostatistician does for a client across an early-phase trial, from the protocol's statistical sections to CSR text. It runs from static files or local R, so client data never leave the analyst's machine; the example studies are synthetic.

- Built a complete worked example of a bioequivalence study: one synthetic trial taken from sample size and randomization through the analysis plan, analysis datasets, pharmacokinetic analysis, tables and report text, all regenerated by a single script.
- Built safety-monitoring tools for first-in-human studies: liver and cardiac screening rules, a Safety Review Committee packet and an interactive dashboard that runs in the browser.
- Built a quality-control harness that recomputes key results a second way and is tested against deliberately planted errors.
- Wrote SAS and R program libraries for standard tables, listings and figures across common early-phase designs, plus simulation studies of two regulatory checks.
- Built four browser-based apps for reviewing clinical trial data (medical coding, de-identification, participant risk and central monitoring) and published their measured limitations.

Public Data Consoles and Coastal Water Resilience

- Built 32 browser-only decision-support consoles on public data feeds and dated snapshots across hazards, public health, access to care and markets; each labels every input as measured, modeled or demonstration.
- Unified my two research methods in Coastal Water Resilience: the ten-metric diagnostic suite from my contamination-resilient cluster randomized trial design, combined with directed source/sink topology accounting from my directed-contagion research, applied to live NOAA, NHC and NWS coastal forecasts across 131 published station locations to drive projected impacts and response plans.

Autonomic Telemetry and Resilience Telemetry

- Built a personal arousal-biofeedback system for work and rest: a multivariate estimate combining heart rate, heart-rate variability, EEG beta/alpha and covariance distance against a personal reference, with prediction-error event detection; live input streams from a Muse S headband and a Hume Band through a local Mac-to-iPhone bridge.
- Applied my directed-contagion research to cross-asset market volatility: spillover networks, transmitter stability and historical replay, tested against held-out baselines that track the loudest and the most persistent assets.

Consilience Engine

- Packaged gene-set analysis as an installable Python library and configurable CLI with offline cache support, building modality-specific graphs from InterPro, Human Protein Atlas, STRING and Gene Ontology sources with permutation-null separation and class-recovery analyses; applied it to a 410-gene nervous-system map and a 132-gene immune replication.

Publications and manuscripts

Submitted manuscripts, with review status as of 26 October 2026:

- **Cluster randomized trial design under spatial interference.** A design framework for trials in which treatment effects spill between sites. *Statistics in Medicine*, submitted. Preprint doi:10.5281/zenodo.23486176.
- **AlphaGenome within-site audit.** What an AI genomics model's variant comparisons actually measure. *Human Genomics*, submitted. Preprint doi:10.5281/zenodo.23543154.
- **A False Floor.** Overturning our own earlier information-limit claim with stronger baselines. PLOS ONE, submitted 22 October 2026. Preprint doi:10.5281/zenodo.23519299.
- **Directed network contagion.** Targeting the units that transmit a contagion versus the hardest-hit ones. *Applied Network Science*, with editor.
- **Consilience Engine.** How different kinds of biological data recognize the same gene groups. *PeerJ*, in peer review.
- **Amygdala invariance.** No evidence for an emotion-specific visual code in the human amygdala. Peer Community In, preprint review. Preprint doi:10.5281/zenodo.21422754.

Published work:

- **Acknowledged contributor** (study-dataset extraction and normalization programs): Osterhage KP, Hser YI, Mooney LJ, et al. Identifying patients with opioid use disorder using International Classification of Diseases (ICD) codes: challenges and opportunities. *Addiction*. 2024;119(1):160-168.
- **Co-author:** Ben-Avie M, Rosenwald L, Depastino R, Iben P. How can we help young adults with autism thrive in the workplace? *The Conversation*, 2016.